National Teachers College
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Bachelor of Science in Information Technology

Final Project Documentation in Computer Programming 2 w/ Lab

StudyWell: Student Wellness & Study Optimizer

UN SDG 3: Good Health and Well-Being

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SECTION 1: INTRODUCTION

1.1 Project Overview & Target SDG

Students commonly experience academic pressure, stress, and difficulty managing their study time, often prioritizing performance at the expense of their mental health. Without a structured way to monitor wellness, these pressures can lead to burnout and long-term health issues. There is a growing need for a system that integrates self-awareness with academic organization, helping students manage their routines in a healthier, more balanced way.

StudyWell is a data-driven wellness and study management system designed to help students monitor their academic health and habits. The system allows users to conduct a "Wellness Check" through a series of evaluative questions covering **Joy of Learning, Time Management, Academic Efficacy, and Perceived Stress.** Based on these metrics, the system provides personalized suggestions, recommended study plans, and historical records to help students maintain productive yet sustainable habits.

This project is directly aligned with **United Nations Sustainable Development Goal (SDG) 3: Good Health and Well-being**, specifically **Target 3.4**, which aims to promote mental health and well-being. By providing a tool for early stress detection and healthy habit formation, StudyWell utilizes technology to address the "preventative" side of global health.

The implementation of this project provides a simple, accessible platform for students to assess their mental state, track their study history, and refine their time management skills. By combining wellness assessment with study optimization, StudyWell promotes self-awareness and organization—ultimately supporting both academic productivity and long-term personal development.

1.2 Problem Statement

For most college students, balancing academic responsibilities, effective time management, and personal well-being is a constant struggle. The pressure to perform academically often comes at the expense of mental health, leading to increased stress and a significant decline in overall academic efficacy. This creates a negative cycle where poor well-being hinders cognitive function, further increasing stress and preventing students from studying effectively.

The primary beneficiaries of this system are college students, specifically those aged 18 to 28, who require a personalized dashboard to manage their academic journey without compromising their health. Recent localized data indicate that Filipino students in this age range exhibit significantly higher levels of psychological distress due to the transition from prolonged lockdowns to in-person learning (Cleofas, 2023). By collecting and analyzing this data, the system directly supports **United Nations Sustainable Development Goal (SDG) 3, Target 3.4**, which focuses on **promoting mental health and well-being through preventative measures.**

Currently, there is a critical lack of accessible tools that allow students to evaluate their wellness in direct relation to their study habits. As corroborated by local research, Klainin-Yobas et al. (2021) state that there is a need for stress management interventions and resilience-based programs to enhance the psychological well-being of college students. However, existing methods for tracking academic progress or mental health often remain separate, preventing students from seeing the correlation between their internal well-being and their external performance. Furthermore, manual tracking is often inconsistent and fails to provide the actionable, data-driven feedback necessary for **early intervention.**

The **StudyWell** system aims to bridge this gap by providing a structured, integrated platform with the following objectives:

- **Structured Assessment:** Allow students to assess their daily wellness through 10 guided questions covering Joy of Learning, Academic Efficacy, and Time Management.
- **Integrated Tracking:** Enable users to log study hours alongside wellness data to monitor long-term health trends.
- **Data-Driven Awareness:** Increase student awareness of their mental state by calculating and displaying real-time scores for overall well-being and perceived stress.
- **Actionable Feedback:** Provide tailored suggestions based on recorded stress levels to support better decision-making and sustainable lifestyle choices.

SECTION 2: REQUIREMENT ANALYSIS

2.1 Functional Requirements (FR)

- **FR1: Automated Data Initialization**
  - Upon startup, the system must retrieve saved records from the “data.txt” file.
- **FR2: Secure Record Management (CRUD)**
  - **Create:** The "Wellness Check Session". Users answer 10 questions to generate a new record.
  - **Read:** The "Study History and Summary". Users can view all days or search for a specific day.
  - **Update:** The "Edit Session". Users can modify a specific day's study hours or re-take the quiz.
  - **Delete:** The "Delete Session". Users can remove a specific day's entry from the collection.
- **FR3: Core Domain Logic (Computation Engine)**
  - Calculate JL, TM, AE, and PS scores and average. It should not just store data, but also process it by recommending a study plan and providing advice based on scores.
- **FR4: Persistent State Storage**
  - When the user selects "Exit" from the menu, the program must loop through your internal vector and overwrite data.txt with the updated data (including any edits or deletions made during the session).
- **FR5: Navigational Menu System**
  - **Main Menu:** (1) Start Wellness Check Session, (2) View Study History and Summary, (3) Edit Session, (4) Delete Session, (5) Study Plan Guide, (6) About us, (7) Sources \ References, (8) Save and Exit to the System
  - **Robustness:** To ensure robustness, the system declared a function int inputValidation(int min, int max); that acts as the input validator for age, choices, and answers to avoid possible loops and crashes.

2.2 Non-Functional Requirements (NFR)

- **NFR1: Robustness & Defensive Programming**
  - **Input Validation:** The program detects and handles type mismatch errors (e.g., entering a letter when a number is expected). Validation mechanisms such as cin.fail() are implemented for clear invalid inputs and prompt the user to re-enter correct data.
  - **Range Validation:** The system only accepts valid numeric inputs within the defined scale (0–4 for questionnaire responses and 0–8 for menu options). Any value outside the acceptable range triggers a re-prompt.
  - **Graceful Failure:** Informative and user-friendly error messages to guide the user when an invalid menu option is selected, a non-existing record is searched or edited, and a file loading fails
  - **Program Stability:** The application continues to operate even after invalid input instead of terminating unexpectedly.
- **NFR2: Structural Modularity**
  - **Multi-File Separation:** The project is divided into multiple files:
    - .h files for class declarations
    - .cpp files for function implementations
    - A separate main.cpp for execution logic
  - **Encapsulation:** All class attributes are declared as private. Access to these attributes is provided through properly defined public getter and setter methods to protect data integrity.
  - **Separation of Concerns:** File handling, computation logic, and menu interface are separated into distinct functions or modules.
- **NFR3: Readability & Professional Documentation**
  - **Naming Conventions:**
    - PascalCase for class names (e.g., WellnessRecord)
    - camelCase for variables and function names (e.g., checkWellness)

- **Consistent Formatting:** Proper indentation, spacing, and bracket alignment are maintained for readability.
- **In-Code Documentation:** Every class and major function includes a structured comment block describing:
  - Purpose
  - Parameters
  - Return values
- **Meaningful Identifiers:** Variable names clearly describe their function and purpose.

- **NFR4: Memory & Resource Efficiency**
  - **Type Selection:** Appropriate data types are used to optimize memory usage: int for counters and scale scores, float or double for average or computed values.
  - **Efficient Collection Usage:** std::vector or arrays are utilized efficiently to manage multiple records without unnecessary duplication.
  - **Scope Management:** Variables are declared in the narrowest possible scope to ensure efficient memory lifecycle management.
  - **Proper File Handling:** File streams (std::ifstream, std::ofstream) is properly opened and closed to prevent memory leaks or data corruption.

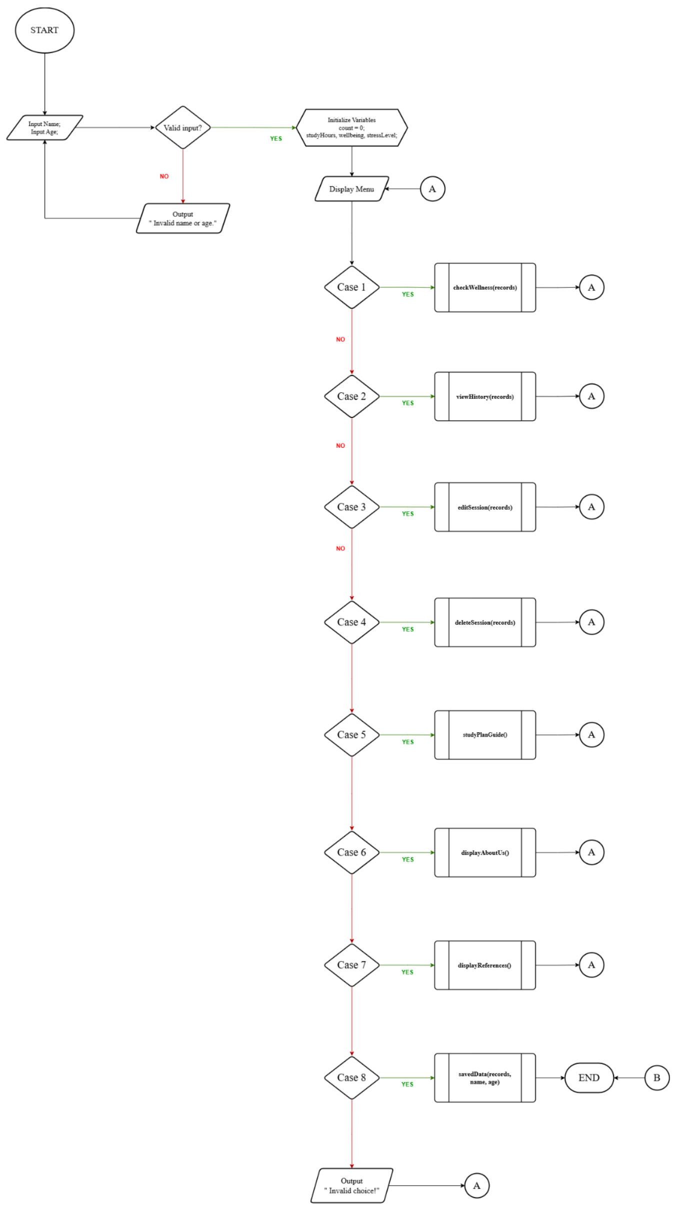
SECTION 3: DESIGN SPECIFICATION

3.1 Algorithmic Flowchart

For better quality, click here → **StudyWell | Flowchart**

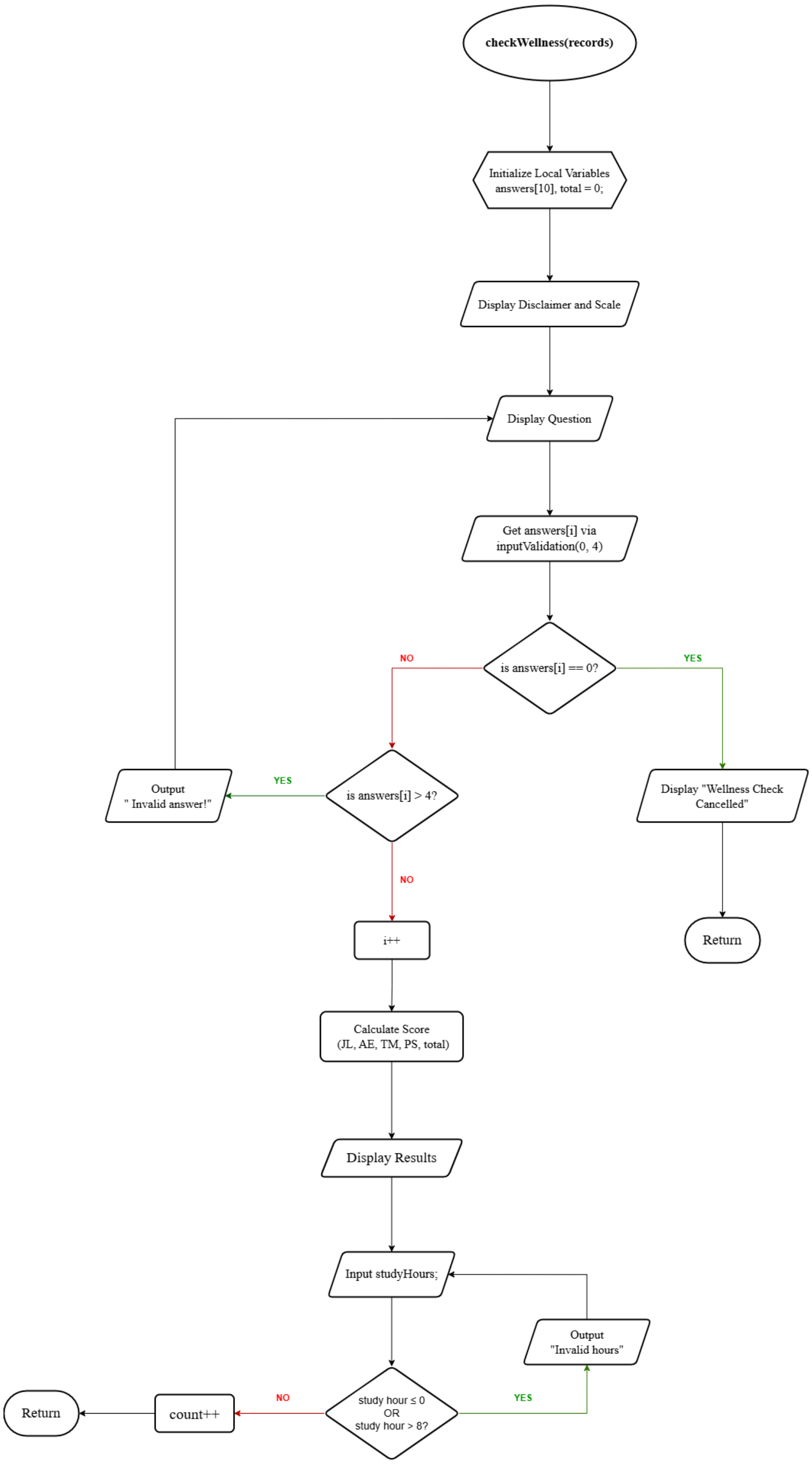
To maximize readability and avoid the visual complexity of a monolithic control-flow diagram, the system's architecture has been systematically broken down into a modular, multi-page layout. The Main Menu serves as the primary structural anchor, mapping out the high-level program execution and user routing using standard Predefined Process blocks.

Main Menu



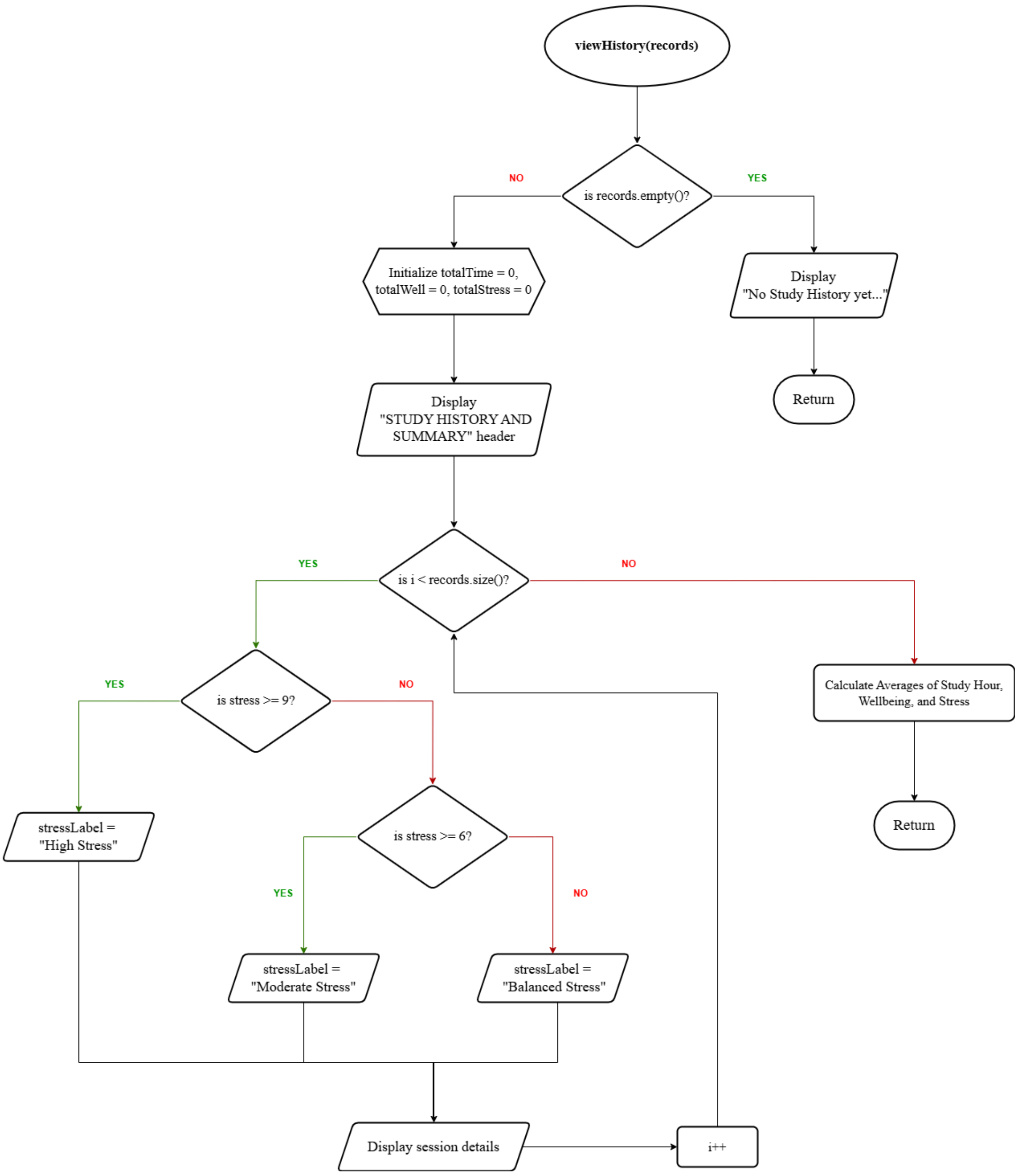
StudyWell-Main Menu.drawio.png

checkWellness(records)



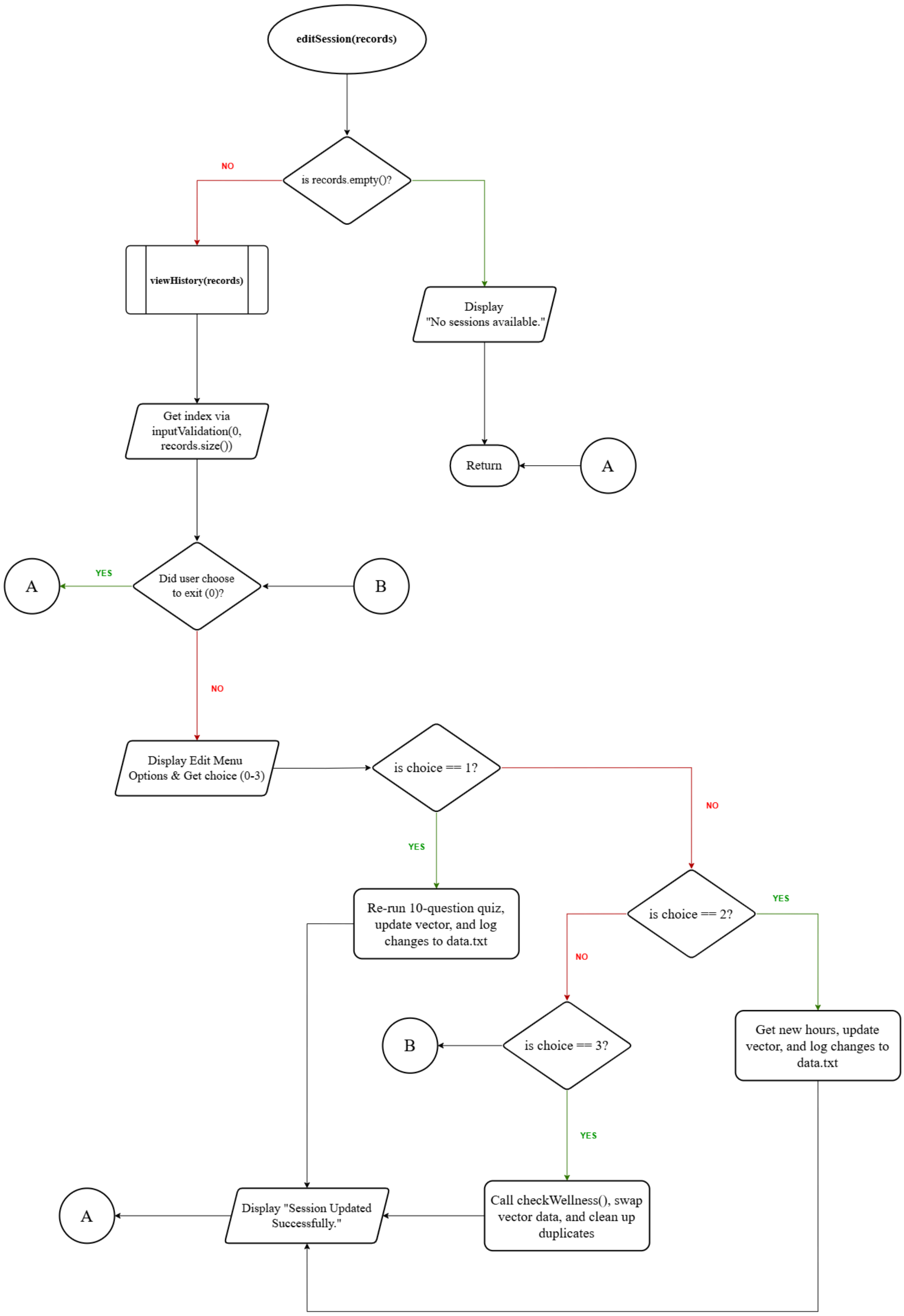
StudyWell-checkWellness(records).drawio.png

viewHistory(records)



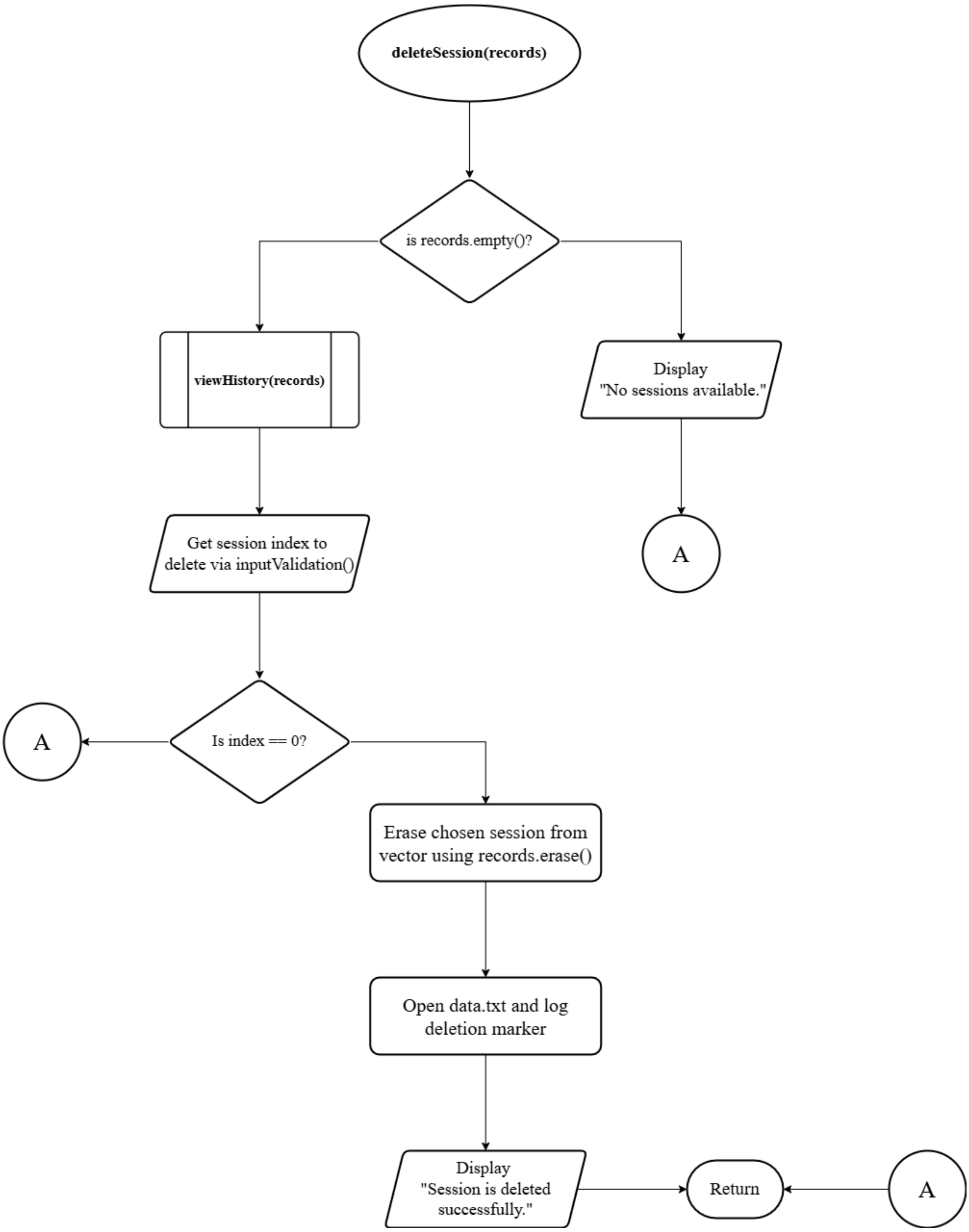
StudyWell-viewHistory(records).drawio.png

editSession(records)



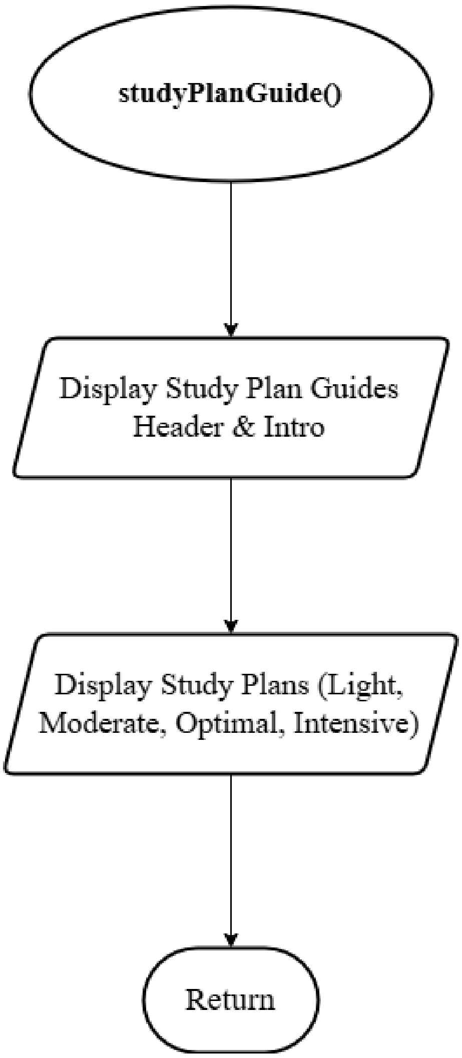
StudyWell-editSession(records).drawio.png

deleteSession(records)



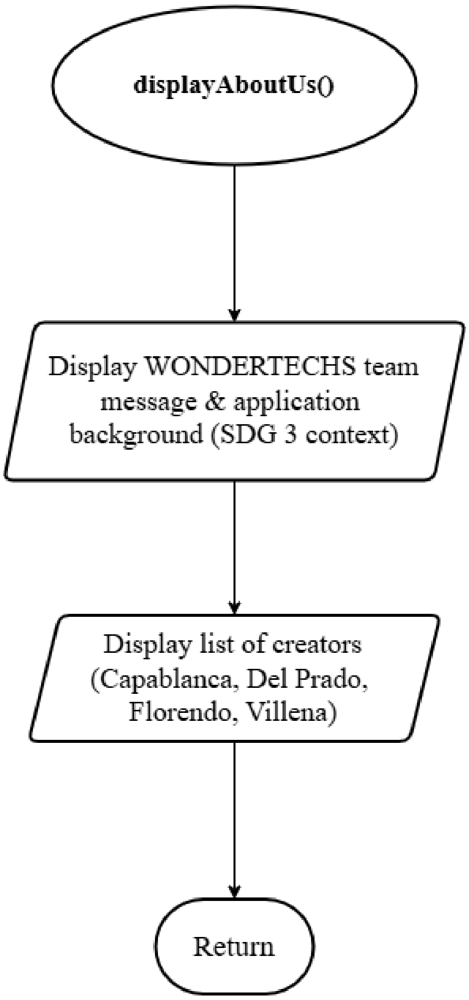
StudyWell-deleteSession(records).drawio.png

studyPlanGuide()



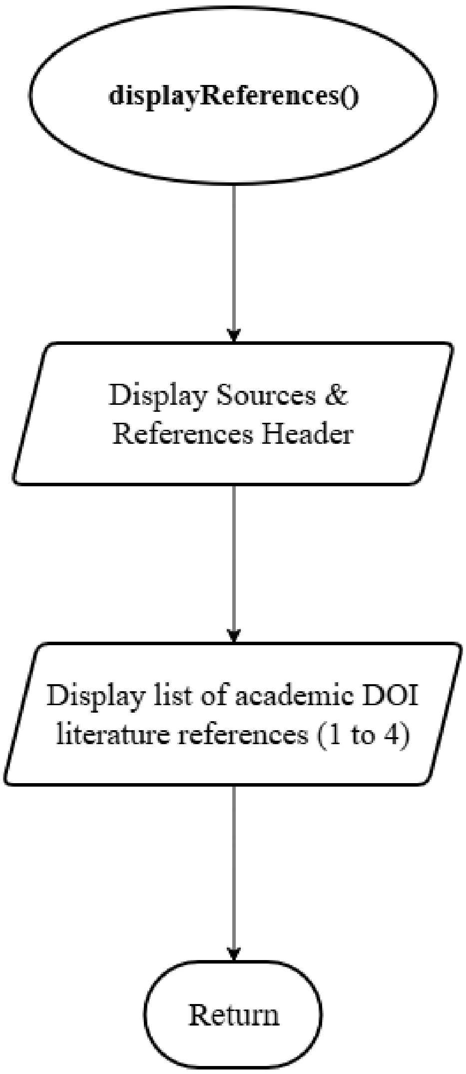
StudyWell-studyPlanGuide().drawio.png

displayAboutUs()



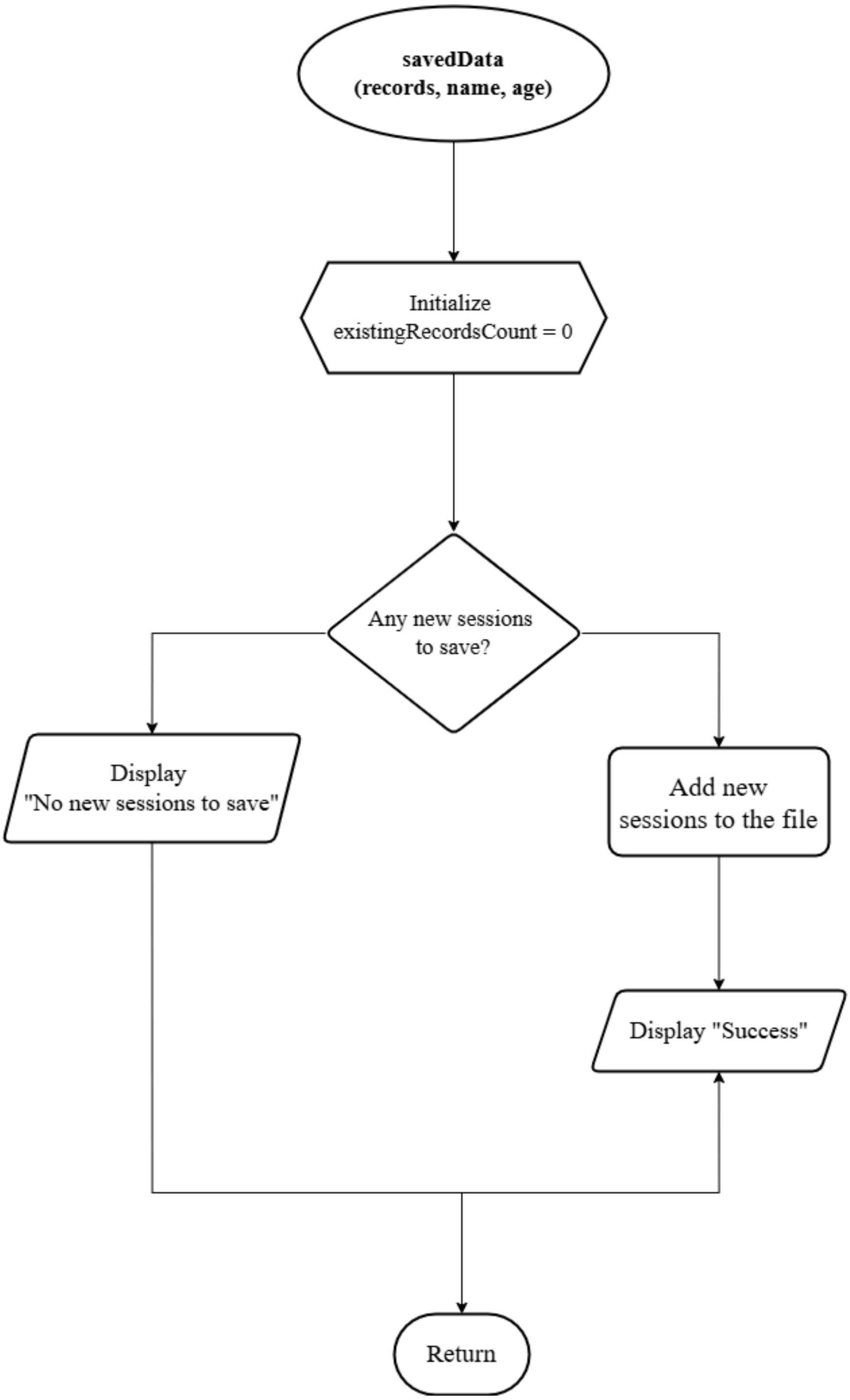
StudyWell-displayAboutUs().drawio.png

displayReferences()



StudyWell-displayReferences().drawio.png

savedData(records, name, age)



StudyWell-savedData(records, name, age).drawio.png

3.2 File Structure Diagram

The application utilizes a flat-file database (`data.txt`) using a Pipe-Delimited Format, where the pipe symbol (`|`) acts as the standard field separator. The system processes two types of file entries

1. **Standard Session Records -** When the user completes a wellness check session, the system logs the metrics sequentially in a single line using this structure:

   Session ID | Study Hours | Wellbeing Score | Stress Score | Study Plan

   Upon system boot, the application reads and parses these lines to rebuild the record collection in memory.

2. **Transaction Log Modifiers -** To modify entries without performing file overwrite operations, the system appends specialized tracking headers to update history:

- **EDITED_MARKER :** Formatted as **EDITED_MARKER | Target Index | New Data**. This informs the program to overwrite the existing data at the specified index with the new values.
- **DELETED_MARKER:** Formatted as **DELETED_MARKER | target Index |0|0|0 null**. This flags a specific session index as removed, prompting the system to skip it during startup mapping and total summary computations.

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